

In this Chapter you will learn about:

- What's beyond our Solar System
- Stars, Galaxies, Milky Way and Black Holes
- Star distances
- The life of Stars (The birth and death of our Sun)
- Looking at Stars

All the students will be able to:

- ✓ Explain the Big Bang Theory of the origin of the Universe.
- ✓ Evaluate the evidence that support the scientific theories of the origin of the Universe.
- ✓ Define the terms star, galaxy, milky way and black holes.
- ✓ Identify bodies in space that emit and reflect light.
- ✓ Describe a star using properties such as brightness and colour.
- ✓ Explain the types of galaxies
- ✓ Explain the birth and death of our Sun.
- ✓ Describe the formation of black holes.
- ✓ Identify major constellations visible at night in the sky.
- ✓ Explain the working of a telescope.
- ✓ Suggest safety methods to use when observing the Sun.



Big, beautiful, barred spiral galaxy NGC 1300 lies some 70 million light-years away. . NGC 1300 spans over 100,000 light-years. It was discovered by John Herschel in 1835. This image was taken by famous Hubble Space Telescope.

WHAT'S BEYOND OUR SOLAR SYSTEM?

- ✓ Explain the Big Bang Theory of the origin of the Universe.
- ✓ Evaluate the evidence that supports the scientific theories of the origin of the Universe.

What is the universe? How did it begin? How big it is? The universe is everything we can see, touch, feel, sense, measure or detect. It includes living things, planets, stars, galaxies, dust clouds, light, and even time. It contains billions of galaxies, and each galaxy contains billions of stars. The space between the stars and galaxies is largely empty.

DO YOU KNOW?

A million is a number where there are six zeros after one like 1,000,000. One thousand counted for one thousand times is called one million. A billion is one million counted for one thousand times. There are nine zeros with one to mention in one billion like 1,000,000,000.

The scientists say that the universe was not always there. According to scientists the universe began to exist about 13.7 billion years ago. The complete size of the universe is difficult to imagine. It is so large that light from very distant objects in the universe travel billions of years before reaching Earth.

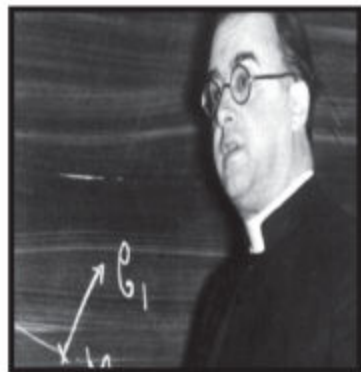
The Big Bang Theory:

The scientific theory that describes the origin of the universe is called the Big Bang Theory. According to this theory:

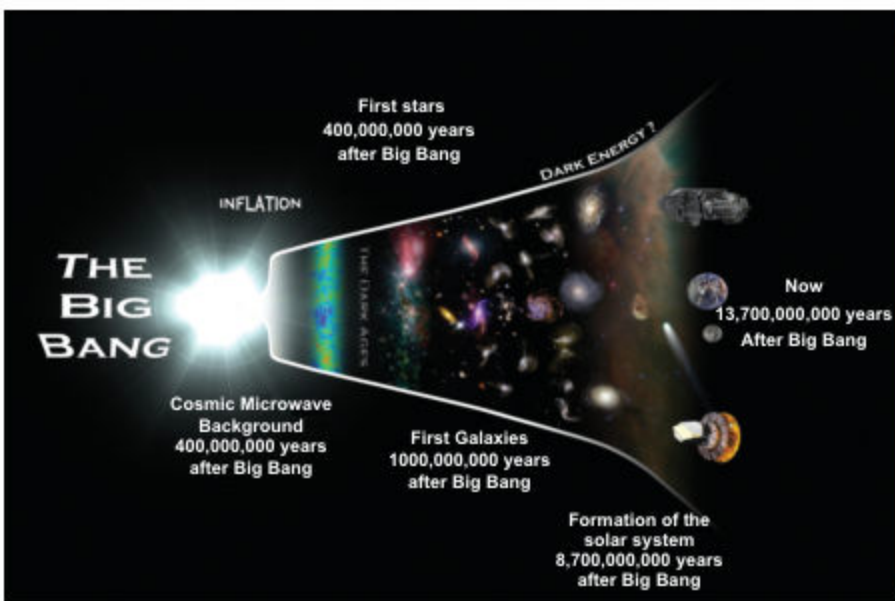
“The universe began to exist as a single point, unimaginably hotter and denser than anything, and it has been expanding ever since.”

Time, space and matter all began with the Big Bang. In a fraction of a second, the universe grew from smaller than a single atom to bigger than a galaxy. It is still expanding today. Due to the expansion of the universe, the conditions in the universe have changed from small to big, from hot to cold and from young to old resulting in the universe we observe today.

In 1917, Dutch astronomer Willem de Sitter used Einstein's theory to describe an expanding universe. Then Alexander Friedmann mathematically proved the idea of expanding universe. In 1920, the Belgian astronomer Georges Lemaitre concluded that if the universe were indeed expanding and has been doing as long as its existence, then there would have been a moment in the distant past when the whole universe occupied a single point. That point and the moment when the Big Bang took place would be the origin of the universe.



Georges Lemaitre



DO YOU KNOW?

One of the interesting things about the universe is that it is currently expanding. It is growing larger and larger all the time. the edge of the universe is also expanding at a faster rate. Scientists say that the edge of the universe is expanding faster than the speed of light.

Fig 12.1: Origin of the universe according to the Big Bang Theory

Evidence of the Big Bang Theory:

Scientists explain that due to expansion of the universe if everything is moving away from each other then it means that the universe is growing larger and larger everyday. By counting backward in time, it is possible to follow that the entire universe was just a point. It is estimated that the universe started growing and expanding about 13.7 billion years ago (age of the universe).

Another evidence supporting the Big Bang theory is the detection of the Cosmic Microwave Background (CMB) radiations.

CMB is said to be “the cooled leftover energy from the hot, early universe that travels freely throughout the universe and still fills space in every direction.”

Scientists consider it as an echo of the Big Bang. Over the time, this light has cooled and weakened considerably; nowadays, we detect it as the microwaves.

In 1992 NASA launched the Cosmic Background Explorer (COBE) satellite. Its purpose was to study the nature of the Cosmic Microwave Background radiations. Instruments on COBE confirmed that the radiation detected by Penzias and Wilson in 1965, perfectly matches the temperature profile of the universe.

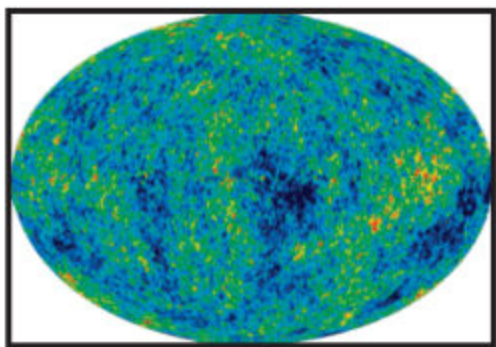


Fig 12.2: All-sky map of CMB

DO YOU KNOW?

Light is the fastest thing in the universe. Light travels with a speed of around three hundred millions meter per second. With this much speed it would take only a second to travel around Earth 7.5 times.



Fig 12.3: COBE Satellite

ACTIVITY 12.1 Model of Expanding Universe

● Material /Resources Required:

Few balloons, pencil and pages, color pencils, gum, scissors

● Procedure:

- 1) Draw the following shapes in multiple on a paper.

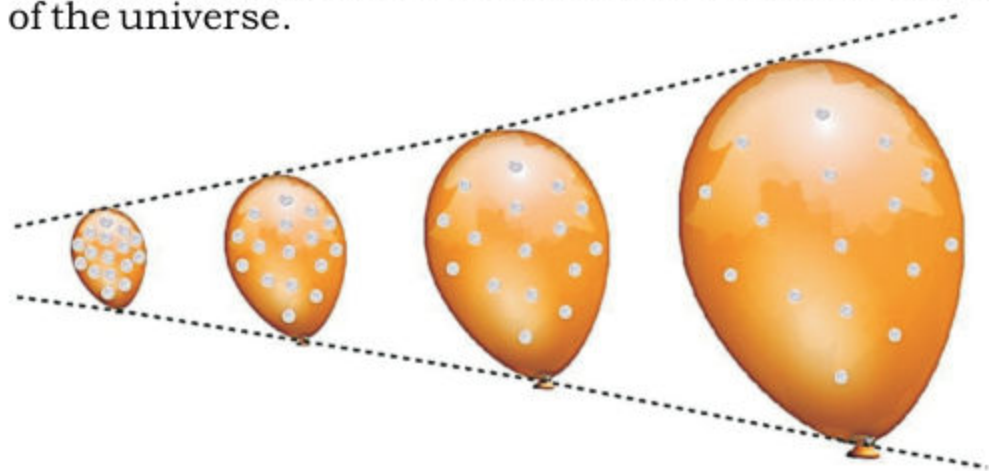


- 2) Cut out all the shapes. Each shape represents a galaxy here.
- 3) Slowly blow up a balloon and stop as soon as the balloon appears round. Hold the end of the balloon to prevent air escaping the balloon.
- 4) With the help of a classmate, paste the shapes you draw earlier on the balloon at various places, covering the whole balloon.
- 5) Now blow up the balloon carefully until it is fully inflated.
- 6) As you do, observe what happens to our galaxies.
- 7) Make different shaped galaxies with different balloons.

Activity Questions:

- 1) Does a galaxy size increase as the balloon expanded?
- 2) What happens to the distances among the galaxies as you blow up the balloon further?

Use these observations to write about important properties of the universe.



STARS, GALAXIES AND MILKY WAY

- ✓ Define the terms star, galaxy, Milky Way.
- ✓ Identify bodies in space that emit and reflect light.
- ✓ Describe a star using properties such as brightness and colour.
- ✓ Explain the types of galaxies.

Stars:

“Stars are huge balls of gas in outer space made from hydrogen, helium and other elements producing light, heat and other forms of energy.”

Our Sun is also a star and is the closest star to Earth. The next closer star to Earth is known as 'Alpha Centauri'. It is about 4.3 light years away from Earth.

On a clear night, a person can see approximately two to three thousand stars. However, with the help of binoculars or telescopes, extremely large number of stars can be observed. We see that some stars are bright in the sky, others are dim. Some look white or blue, while others look red or orange. Do you know color and brightness of stars indicate their interesting properties?

DO YOU KNOW?

Light year is a unit of distance and not the time. If one travels in a direction with the speed of light for one year (365 days), then the distance covered will be equal to one light year.

Colours of stars:

Stars appear red, orange, yellow and blue. Colours of stars are due to difference in temperature they possess. The coolest stars are the red stars and their temperature is around 3,000 degrees Celsius. The Sun has temperature of around 6,000 degrees Celsius and glows orange/yellow. The blue stars, which are the hottest, having temperature more than 25,000 degrees Celsius.

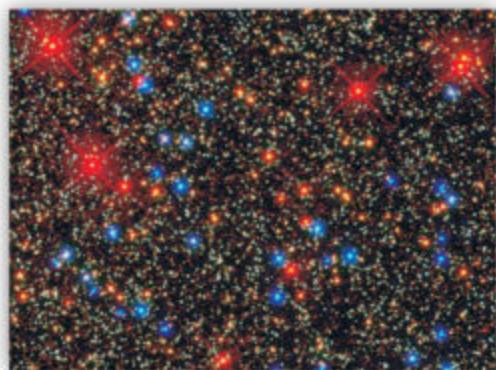


Fig 12.4: Colours of stars; blue stars are hotter than red stars

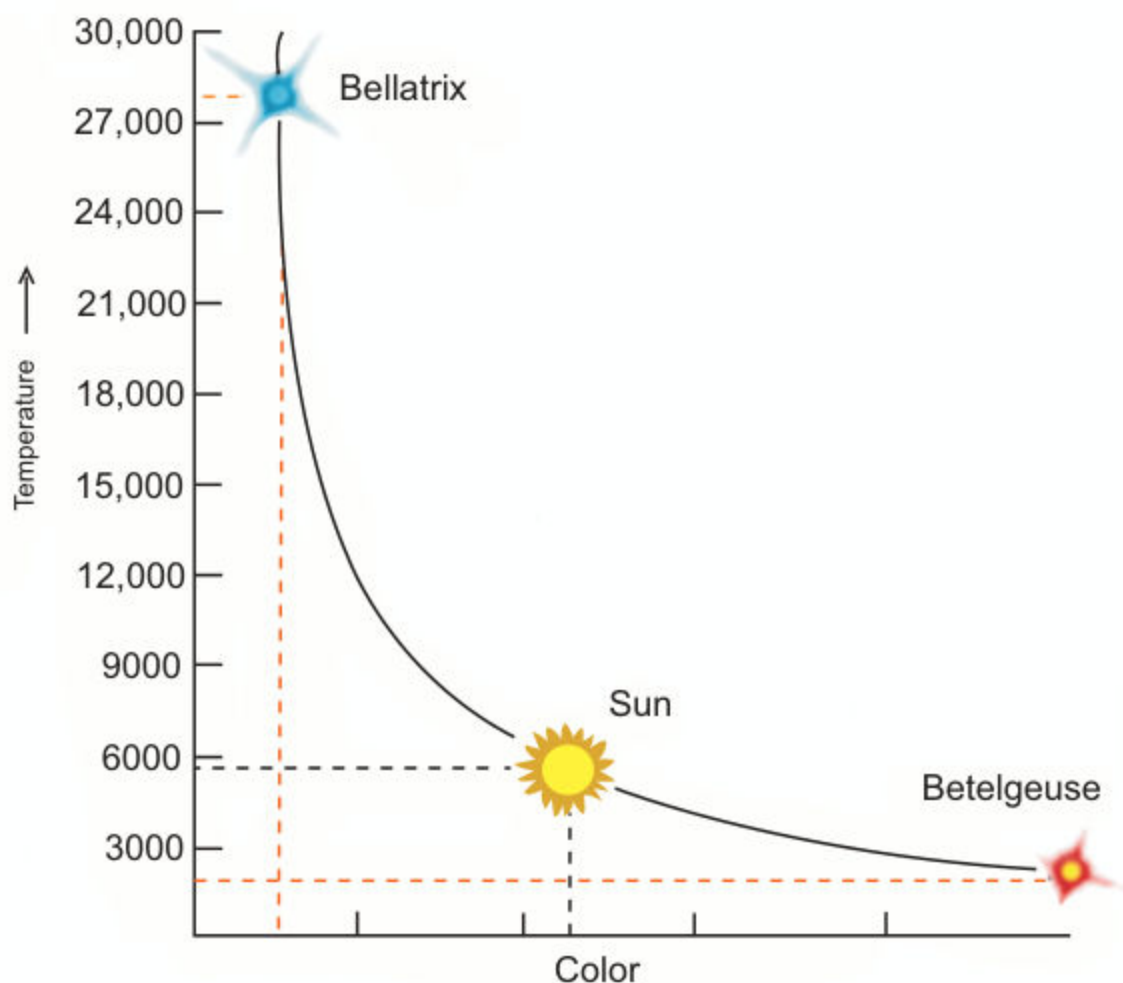


Fig 12.5: The graph showing the change in colour as the temperature of a star changes

Table below shows the colour and temperature of some brightest stars in the sky.

Colours and Temperature of Stars		
Star Name	Color	Temperature ($^{\circ}\text{C}$)
Betelgeuse	Red	3000
Arcturus	Orange	4000
Sun	Yellow	6000
Polaris	Yellow	5800
Vega	White	10000

Brightness of stars:

Stars vary in their brightness levels as seen from Earth. A hot star gives off more energy than a cooler star. Does this then mean that a hot star is going to appear brighter than a cooler one in the sky? The answer to this depends on a few factors. These include:

The size of a star:

If two stars having the same temperature but differ in size then the larger star gives off more light than the smaller star; and will appear brighter in the sky.

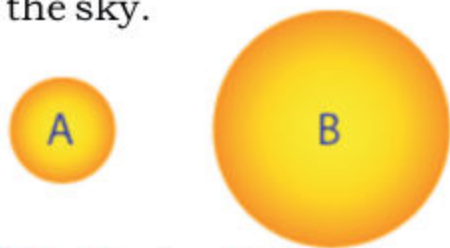


Fig 12.6: Star A and B have the same temperature.
B is bigger than A and gives off more light than A

The distance to the star:

All the stars we see in the night sky are at vast distances but some are much closer as compare to others. For two stars of the same size and temperature, the closer one will appear brighter than the star which at more distance. An analogy is a row of street lights; the closer ones appear much brighter than those at some distance.



Fig 12.7: Apparent brightness of street light changes from nearest to farthest

Intervening matter:

Space is not a perfect vacuum. Dust and gas between stars can absorb and scatter starlight leading to a reduction in brightness.

Galaxies and Their Types:

“Galaxy is a vast collection of stars, gas, dust and other forms of matter which are bound together gravitationally as a unit”.

It is estimated that there are 50 to 100 billion galaxies of different shape and kind in the universe.

Each galaxy has its own identity and age. The earth is part of a galaxy, called the Milky Way.



Fig 12.8: A small portion of the sky as seen through Hubble Space Telescope; every bright spot in this figure is a galaxy

Types of galaxies:

In the 1920s the pioneering astronomer Edwin Hubble, who devoted his life to studying galaxies, proposed a way to classify galaxies based on their shapes. This sequence is known as the Hubble sequence.

“According to this sequence galaxies are generally grouped by their appearance into spiral, elliptical, lenticular and irregular”.

Galaxies are also often categorized by the characteristics they possess other than their appearance. For example, there are starburst galaxies, merging galaxies, active galaxies, radio galaxies and many more.



Sir Edwin Hubble

Elliptical galaxy:

An elliptical galaxy appears to be an oval shaped from our point of view as seen from the telescope. Astronomers have found that elliptical galaxies can have length, width and height that are all different from one another. Elliptical galaxies can be shaped like gigantic basketballs, ostrich eggs or anything like that. Due to their yellowish appearance, elliptical galaxies mostly contain old stars.



Fig 12.9: Two Elliptical galaxies NGC 3923 & and IC 2006

Spiral galaxy:

A spiral galaxy appears to have spiral-shaped structures or arms that contain bright young bluish stars. Spiral galaxies also have oval-shaped bulges at their centers which are filled with stars. There is also a thin disk of spinning gas surrounding the bulge; and a thinly populated halo that encloses both the disk and the bulge.



Fig 12.10: Whirlpool galaxy and NGC 1566:
two beautiful spiral galaxies

Lenticular galaxy:

A lenticular galaxy is a lens-shaped galaxy that has components of both elliptical and spirals galaxies. It can be viewed as either an elliptical galaxy with a disk surrounding its outer edge, or as a spiral galaxy with a very large bulge and almost no spiral arm structure.



Fig 12.11: A lenticular galaxy

Irregular galaxy:

An irregular galaxy does not fit well into the standard categories of elliptical or spiral galaxies. Two examples of irregular galaxies are the large magellanic cloud and small magellanic cloud, which are visible from Earth's southern hemisphere.



Fig 12.12: Large magellanic cloud; an Irregular galaxy

The table below lists some of the well-known galaxies.

Some Well-Known Galaxies		
Common Name	Catalog Name	Galaxy Type
Andromeda Galaxy	Messier 31	spiral
Messier 87	NGC 4486	elliptical
Pinwheel Galaxy	Messier 101	spiral
Sombrero Galaxy	Messier 104	lenticular
Whirlpool Galaxy	Messier 51	spiral
Large and Small Magellanic Cloud	-	Irregular

The Milky Way Galaxy:

The Milky Way is the galaxy we live in. It contains the Sun and at least one hundred billion other stars. There are billions of tons of free floating clouds of gases with dust and several hundred star-clusters in milky way.

What is the shape of Milky Way? For us figuring out the shape of the Milky Way is somewhat like a fish trying to figure out the shape of the ocean. Based on observations, it appears that the Milky Way is a spiral galaxy. The disk of the Milky Way is about 100,000 light-years across and 1,000 light-years thick. Earth and the Sun are about 25,000 light-years away from the center of Milky Way. Our Solar System is moving through the Milky Way's disk in roughly circular orbit around the center of the galaxy and complete one orbit in around 250 million years.

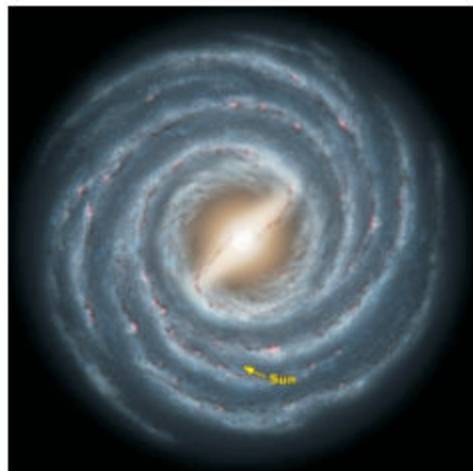


Fig 12.13: Possible shape of Milky Way galaxy; the arrow marks the position of our Sun in the galaxy

Activity 12.2 Model of Milky Way Galaxy and its different parts

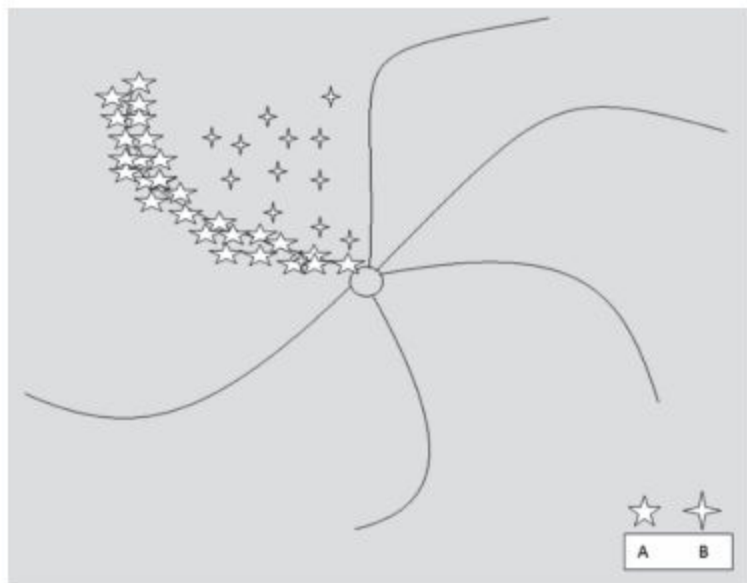
Material / Resources Required:

A drawing sheet, a black sheet, color markers, pencil and pages, tissue papers, gum, glitter

Procedure:

- 1) Mark the center of drawing sheet.
- 2) Make a ball of paper and paste glue over it. Now take few tissue papers and wrap the ball with it such that the tissue papers are firmly glued to the ball. Leave it to dry.
- 3) After it is dried, color the ball with orange and yellow markers.
- 4) Paste the ball at the center of the sheet you marked earlier. The ball is the center of galaxy.
- 5) Now take pencil and draw different spiral arms around the center of galaxy as shown below in diagram.
- 6) Draw and cut out lots of stars as shown below A and B. Glue A-stars to each spiral arm of the galaxy while use B-stars to fill in the gap between the arms.
- 7) When all the stars are firmly dried and stuck to the sheet,

- 8) Use blue glitter to fill in the remaining white space surrounding the spiral arms.
- 9) Make a small paper ball and color it yellow.
- 10) Paste this ball at an approximate location of the Sun as shown in the Figure 12.13. Mark the extent of the disk in light years with an arrow.



Star distances:

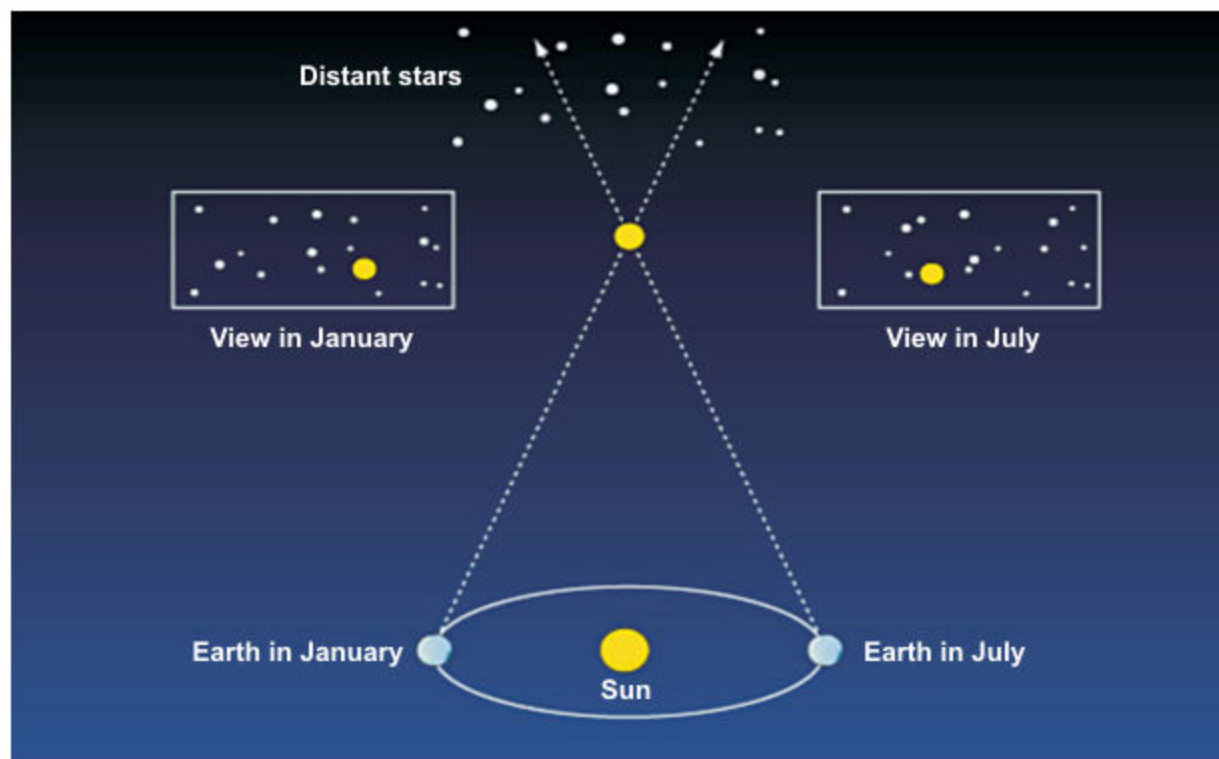
The determination of distances to stars is important in understanding their physical nature. Astronomers have developed a variety of clever techniques for measuring the vast distances to stars, one is called Parallax.



Fig 12.14: Change in the background as seen from two different positions

See figure (12.14), as the observer moves between the two positions, he would see the same tree, but it would appear to move against the background.

This effect can be used to measure the distances to nearby stars. As Earth orbits the Sun, a nearby star will appear to move against the more distant background stars, in the same way the tree appears to move against the more distant mountains in the diagram above. Astronomers can measure a star's position once and then again 6 months later and calculate the apparent change in position. The tiny shift in star's position in the sky is called its parallax.



Since this universe is so big, distances are no longer measured in meters or kilometers. Such distances to nearby stars are measured in 'light year' unit. The brightest star visible in the sky without a telescope is Sirius which has a distance of little more than 8 light years. It is interesting to note that light reaches us from the Sun in about 8 minutes and from Sirius in about 8 years.

DO YOU KNOW?

Effect of parallax can be observed very easily. Hold your hand out in front of you and look at it with your left eye closed, then your right eye closed. Your hand will appear to move against the background.

THE LIFE CYCLE OF STARS

- ✓ Explain the birth and death of our Sun.
- ✓ Describe the formation of Black Holes.

How long a star will live and how it will die, depends upon its mass at the time of its birth. Low mass stars have different endings than the violent and explosive ending of high mass stars. Our Sun is an average type low mass star among billions of other stars in our galaxy.

Scientists believe the Sun is about half way through its lifetime. The Sun and all the planets of our Solar System began from a huge cloud of gas and dust about 4.5 billion years ago. These clouds are known as nebulae.

Proto-star phase:

Nebulae are made up of 97% hydrogen and 3% helium. Then something caused the cloud to contract under its own gravitational forces and a ball of gas and dust is formed around which the material of gas and dust circled. At this time it is known as protostar. "Proto" means "early" or "before." So a protostar is the first step in becoming a full-fledged burning star.

Main Sequence phase:

The Sun spent about 100,000 years as a proto-star before the nuclear fire began at the core. Then it settled down into its current form. The most stable part of its life known as "the main sequence". Generally, a star remains in this phase for most part of its lifetime burning hydrogen.

A star leaves its main sequence phase when it runs out of hydrogen and starts fusing helium and other elements. In 3.5 billion years from now, the Sun will be 40% more brighter than it is right now.

Red-Giant phase:

About 4.5 billion years from now the Sun will exit the main sequence. With its hydrogen exhausted in the core, the helium ash will become unstable and collapse under its own weight and start heating the core. The Sun will then grow in size and will become the Red Giant.

Planetary-Nebula phase:

Then about 250 million years the Sun will become unstable and begin losing its mass. After 500,000 years or more, only half of the Sun's current mass will remain and its outer layers will begin to form a planetary nebula.

White Dwarf:

The core of the Sun will cool towards a white dwarf. The planetary nebula will disperse in about 10,000 years, but the white dwarf will survive for trillions of years before fading to black.

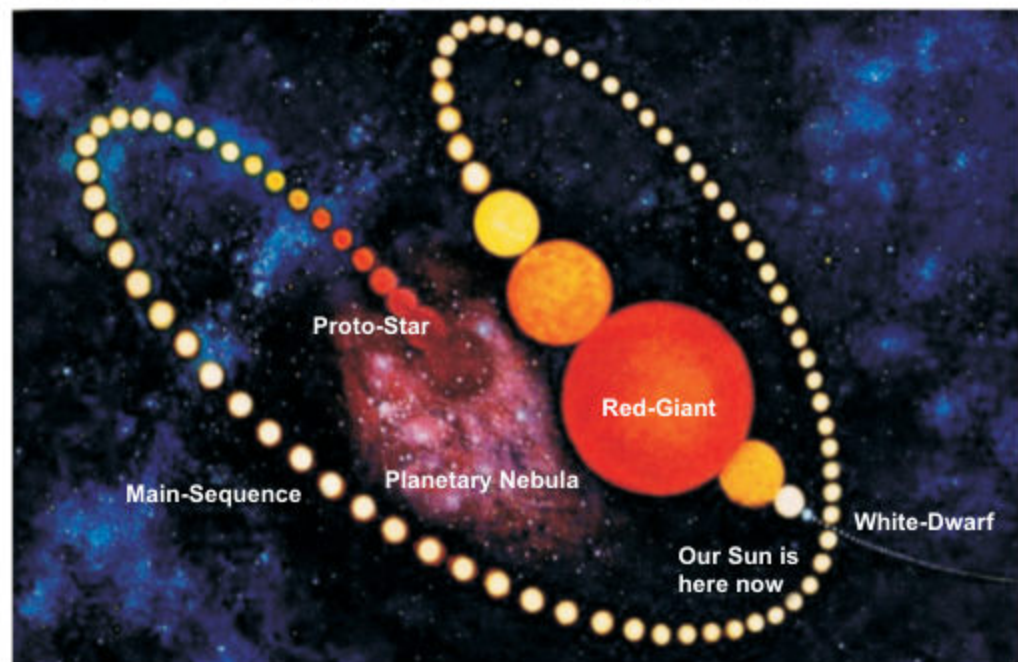


Fig 12.15: Life cycle of Sun-like stars

Black Holes and their Formation:

“A black hole is a region of space where matter has collapsed in on itself”.

This catastrophic collapse results in a huge amount of mass being concentrated in an incredibly small area. The gravitational pull of this region is so great that nothing can escape not even light.



Fig 12.16: An artist conception of a black hole

A black hole does not have a surface, like a planet or star. They are invisible.

Black holes are formed when a heavyweight star, about 10 times heavier than the Sun, ends its life in a supernova explosion. What is left of the star collapses into an area only a few kilometres across. A supernova is an exploding star that blasts parts of the star into space.

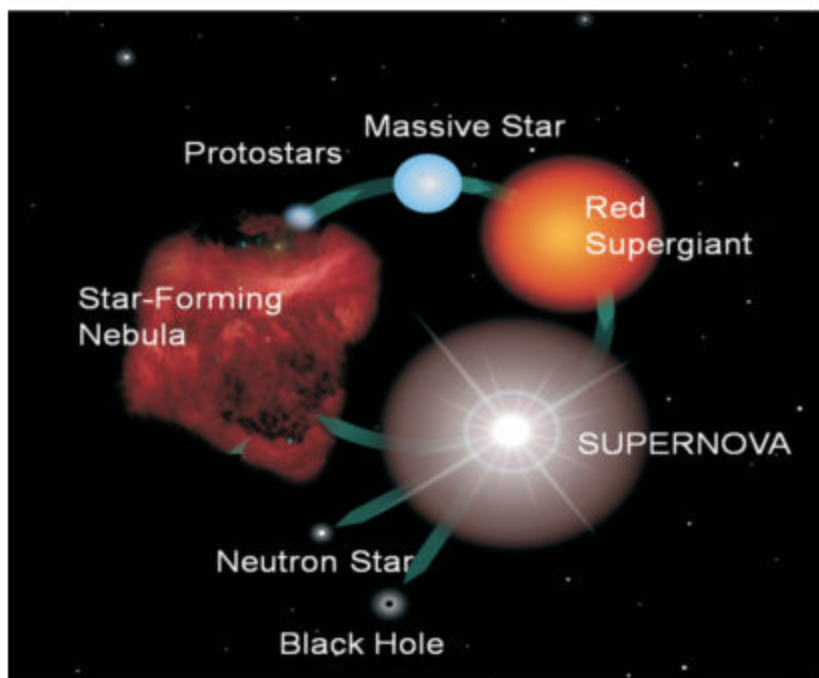


Fig 12.17: Life cycle of a massive star and formation of black hole

LOOKING AT STARS

- ✓ Identify major constellations visible at night in the sky.
- ✓ Explain the working of a telescope.
- ✓ Suggest safety methods to use when observing the Sun.

Constellations

A constellation is a group of stars in the sky that, when viewed from earth, create an outline of some recognizable shape or pattern. Modern constellations are mostly named after mythological themes such as gods, legendary heroes, creatures or structures. Although most constellations resemble the figures after which they are named, others are not as recognizable.

The constellations encompass the entire sky and only provide a visual reference for objects in the sky. There are a total of 88 constellations in the sky which are internationally recognized. Table below lists some well-known constellations.

Well-known Constellations		
Name	Common Name	Famous stars in the Constellation
Aquila	The Eagle	Altair
Bootes	The Hunter	Arcturus
Canis Major	The Big Dog	Sirius
Gemmini	The Twins	Castor, Pollux
Leo	The Lion	Regulus
Orion	The Hunter	Rigel, Betelgeuse, Bellatrix
Taurus	The Bull	Aldebran
Ursa Minor	The Little Bear	Polaris

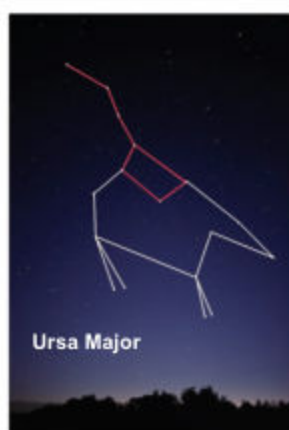
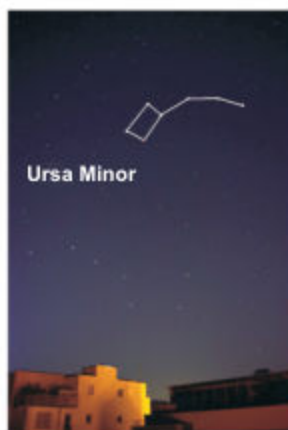
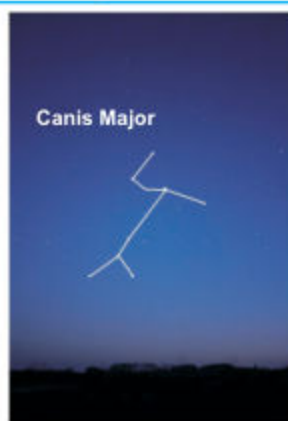


Fig 12.18: Constellation listed in table above as they can be seen in the sky. Lines are marked connecting stars which outlines the shape of the constellation

Activity 12.3 Making a Constellation Map

Material / Resources Required:

Drawing paper, black marker, cardboard, pencil, gum

Procedure:

- 1) Make 10 to 15 stars on cardboard and color them white and cut each star out from the cardboard.
- 2) Paste the stars on the drawing sheet as shown in the constellation figures of Orion and Ursa Minor.
- 3) Join the lines between the stars to make the pattern.
- 4) Now ask the student to think about some imaginary objects which they could make with the help of stars like hen, duck, bear etc.



How does a telescope work?

Telescope:

A telescope is an optical instrument used to see far remote objects like stars and planets.

Telescope gathers light from distant sources in such a way that an image can be produced. The first type of telescopes was made with glass lenses attached to handheld cylinders or tubes. Today, telescopes are made in many different ways, and used together with all kinds of scientific instruments to study the universe near and far.

There are two main types of telescopes: a refractor, which uses lenses to collect light, and a reflector, which uses mirrors.

Most refracting telescopes use two main lenses. The bigger lens is called the objective lens, and the smaller lens used for viewing is called the eyepiece lens. Refracting telescopes work as their objective lenses gather more light coming from an object than a human eye can collect on its own and focuses it to a point. Then the eyepiece lens magnifies image of the object which can be seen by viewer as brighter, clearer and magnified image.

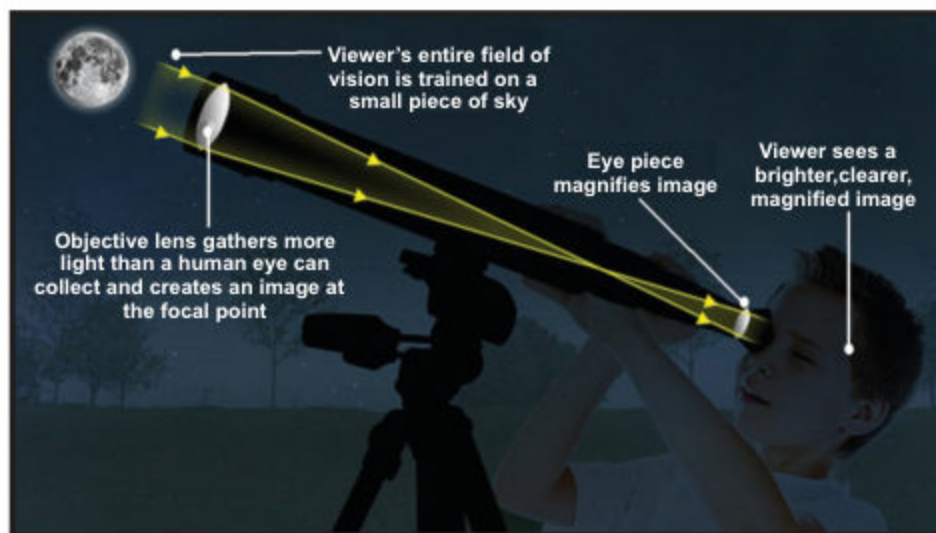


Fig 12.19: Working of a refracting (lenses) telescope

Reflecting telescopes have curved mirrors which served to collect incoming light from objects in the sky and are called objective mirrors. It bends light and makes parallel light rays converge to a focus. However, there is a secondary mirror in the telescope which directs the incoming focus light rays to eyepiece from where objects can be seen.

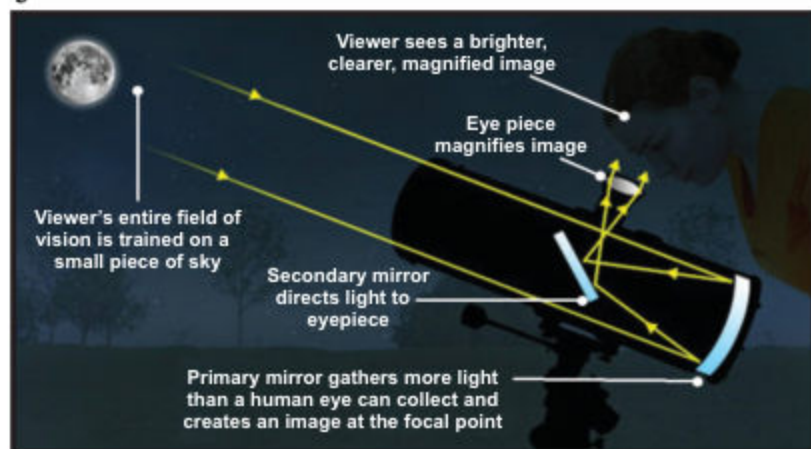


Fig 12.20: Working of a reflecting (mirror) telescope

Safety Methods when observing the Sun:

The Sun is so bright that prolonged, direct exposure can cause permanent damage to the retina, leading to loss of vision or blindness. To observe the Sun safely, more than 99% of the Sun's light must be filtered before it reaches eyes. One should never view the Sun with the naked eye or with any unfiltered optical device, such as binoculars, telescope, regular sun glasses etc.



Special filters should be used to cover up the objective lens of the telescope.

Galileo, the scientist who invented the telescope, looked at the Sun through a telescope, suffered permanent eye damage.

One safe way to observe the Sun is to project an image of the Sun through a telescope or binoculars onto a white screen or any other plain surface. This is an indirect way of observing the Sun.

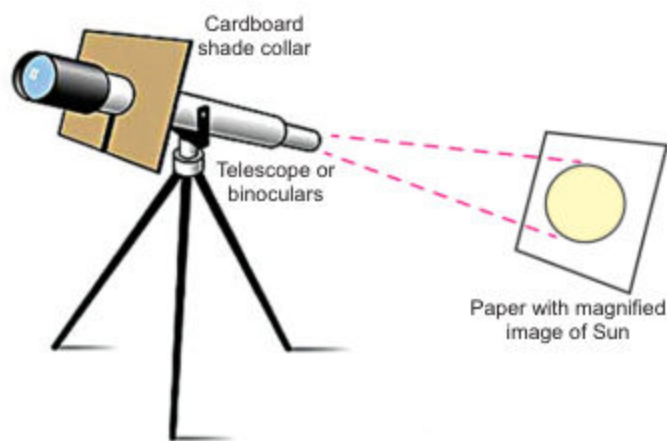


Fig 12.21: Projection method for observing the Sun



Fig 12.22: Special solar glasses for observing the Sun

Summary

- The scientific theory that describes the origin of the universe is called the Big Bang Theory.
- The universe began to exist as single point and it has been expanding ever since.
- The most convincing evidence confirming the Big Bang theory is the Cosmic Microwave Background (CMB) radiations.
- Stars are huge balls of gas made from hydrogen, helium and other elements producing light, heat and other forms of energy.
- Colours of stars are due to their temperature. Blue stars are hotter than red and orange stars.
- A galaxy is a vast collection of stars, gas, dust and other form of matter which are bound together gravitationally as a unit.
- Galaxies are generally grouped by their appearance into spiral, elliptical, lenticular and irregular.
- Parallax is the apparent displacement of an object because of a change in the observer's point of view.
- Life stages of a star are Proto-star, Main Sequence, Red-Giant, Planetary-Nebula and White Dwarf
- Black hole is a region of space where matter has collapsed in on itself.
- A constellation is a group of stars in the sky that, when viewed from Earth, create an outline of some recognizable shape or pattern.
- Refractor telescope uses lenses to collect light and a reflector telescope uses mirrors.
- The Sun emits dangerous radiation. Safety measure must be taken before viewing the Sun.

Review Exercises

1. Fill in the blanks with appropriate words given below.

hydrogen, spiral, higher, matter, stars, elliptical,
dust, light, Origin, helium, gas,

- i) The scientific theory that describes the _____ of the universe is called the Big Bang.
- ii) Stars are huge balls of gas mainly consist of _____ and _____.
- iii) Blue stars have _____ temperature than orange and red stars.
- iv) A galaxy is a vast collection of _____, _____ and _____.
- v) A lenticular galaxy is an intermediate form of _____ and _____ galaxies.
- vi) Black hole is a region of space where _____ has collapsed on itself.
- vii) A telescope is an instrument that gathers _____ from distant sources.

2. Circle T for True and F for False state. Provide a correct statement for the False Statement.

- i) The universe is 10 billion years old.
- ii) CMB stands for Cosmic Matter Blue radiations.
- iii) NASA launched COEB satellite in 1992.
- iv) The Sun has temperature of around 6000 degree Celcius.
- v) The named "Arcturus" is a hot blue star.
- vi) Earth galaxy is called "Milky Way".
- vii) Einstein proposed a way to classify galaxies.
- viii) The Sun is about 15000 light years away from the center of our galaxy.
- ix) The final stage of a Sun-like star is a black hole.
- x) Constellations provide only a visual reference for the objects in the sky.

3. Write down the short answers to the following questions.

- i) What type of a star ends its life cycle as a black hole?
- ii) What is a light year?
- iii) Define the term “constellation”.
- iv) On what factors does the brightness of a star depend?
- v) What are Cosmic Microwave Background radiations?

4. Write down the detailed answer of the following questions.

- i) Explain the Bing Bang theory about the origin of the universe.
- ii) Explain the stages of life-cycle of a low mass star.
- iii) Describe the types of galaxies.
- iv) How do we measure distances to nearby stars.
- v) Explain the birth and death of our Sun.